

Embedded Linux & System Software Interview Handbook

Chapter 6 - Device Tree & Platform Drivers (Part 1)

Topics: Device Tree Fundamentals, DTS/DTB, Platform Drivers, Driver Matching

6.1 Why Device Tree?

Modern ARM-based Linux kernels separate hardware description from kernel code using the Device Tree. This allows a single kernel image to support multiple boards without recompilation.

Boot Flow

BootROM → SPL → U-Boot → Kernel + DTB → Driver Matching → probe()

6.2 Important Files

File	Purpose
.dts	Board description
.dtsi	Shared include file
.dtb	Compiled binary tree
dtc	Device Tree Compiler

6.3 Basic DTS Structure

```
/ {
model = "Example Board";
compatible = "vendor,board";
memory { };
cpus { };
soc {
uart@1000 {
compatible = "vendor,uart";
status = "okay";
};
};
}
```

6.4 Platform Drivers

Platform drivers are widely used for on-chip peripherals such as UART, GPIO, I2C, SPI, watchdogs and timers. The kernel matches the compatible string in the Device Tree with the driver's of_device_id table.

Matching Process

Device Tree compatible → of_match_table → platform_driver → probe()

Typical probe() Responsibilities

- Obtain resources
- Map MMIO registers
- Enable clocks
- Configure GPIOs
- Request IRQs
- Register the device

Common APIs

API	Purpose
platform_driver_register()	Register driver
platform_get_resource()	Get MMIO resource
devm_ioremap_resource()	Map registers
devm_request_irq()	Register interrupt
devm_kzalloc()	Managed allocation

Debugging Tips

If probe() is never called, verify the compatible string, ensure the DTB loaded by U-Boot matches the running board, inspect dmesg, and confirm the device node has status="okay".

Interview Questions

1. Why was Device Tree introduced?
2. Difference between DTS, DTSI and DTB?
3. How does Linux match a platform driver?
4. What is the purpose of compatible?
5. Why would probe() never execute?

Key Takeaways

Device Tree is the bridge between hardware and the Linux kernel. Understanding platform drivers and the matching process is essential for BSP and embedded Linux interviews.